



VIRGINIA COMMONWEALTH UNIVERSITY

ASSIGNMENT REPORT

Exploratory Data Analysis of the Dry Bean Dataset

Course Code	MBAV 443M
Course Title	Statistical Modeling and AI-Driven Analytics
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Student Name	Neha Maria Jesluck
Student ID	2528708
Program	MBA (Marketing)

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1. Introduction

The Default of Credit Card Clients Dataset consists of credit card holder information, which includes demographics, credit limits, bill statements, payment history and default status. This dataset is commonly used in data analytics and machine learning for analyzing credit behavior and predicting credit card defaults. The purpose of this assignment will be to perform Exploratory Data Analysis (EDA) using Python on this dataset. EDA is used to discover the structure and attributes of the data, assess quality and to find patterns, identify anomalies and to evaluate relationships among variables. The findings from EDA can provide the basis for future predictive modelling and credit risk assessment.

2. Data Understanding and Quality Assessment

The dataset contains customer information along with financial and payment-related attributes. Variables include demographic factors such as gender, education level, marital status, and age, as well as financial indicators such as credit limit, bill amounts, payment amounts, and repayment history. Data quality assessment was conducted using functions such as `info()`, `dtypes`, `isnull()`, `isna()`, and `uplicated()`. These techniques helped verify the consistency and completeness of the dataset.

The dataset was examined for:

- Data dimensions and structure
- Data types of variables
- Missing values
- Duplicate records
- Unique values and category distributions

Missing value analysis confirmed that the dataset contained minimal or no missing values, making it suitable for analysis. Duplicate records, if present, were identified and removed using the `drop_duplicates()` function to improve data integrity. The overall data quality was found to be satisfactory, requiring only limited preprocessing before analysis.

3. Methodology

The analysis was performed using Python in a Jupyter Notebook environment. The following libraries were utilized:

- Pandas
- NumPy
- Matplotlib
- Seaborn

The methodology followed these steps:

1. Loading and inspecting the dataset
2. Understanding dataset dimensions and structure
3. Verifying data types and data quality

4. Checking for missing values and duplicate records
5. Performing descriptive statistical analysis
6. Exploring categorical variables through frequency analysis
7. Conducting group-wise analysis using GroupBy and aggregation functions
8. Evaluating relationships between variables using correlation analysis
9. Creating visualizations to understand distributions, trends, and patterns
10. Interpreting findings and drawing conclusions

Descriptive statistics including mean, median, mode, standard deviation, variance, minimum, maximum, and quantiles were calculated to summarize numerical variables.

4. Visualization

Several visualization techniques were employed to better understand the characteristics of the dataset. Histograms and Kernel Density Estimation (KDE) plots were used to examine the distribution of numerical variables such as credit limits and bill amounts. Boxplots and violin plots helped identify variability and potential outliers within the data. Count plots were used to visualize the frequency distribution of categorical variables such as gender and default payment status. Bar plots were used to compare average credit limits and other financial indicators across different customer groups. Scatter plots and joint plots were utilized to explore relationships between variables such as credit limits and bill amounts. Pair plots enabled simultaneous visualization of relationships among multiple variables. Correlation heatmaps provided a graphical representation of the relationships between numerical features, highlighting strong positive and negative correlations among financial variables. These visualizations collectively provided valuable insights into customer behavior, spending patterns, and default tendencies.

5. Key Insights

Several important findings emerged from the exploratory analysis. The first finding is that the customer credit limits vary substantially within the available data, and as a result, there is considerable credit exposure across the customer base. The descriptive statistics also indicate that there is great variability between different customers' financial indicators (i.e. the amount of their bills and the amount of their payments). In addition to the findings regarding the credit limits and variability in the financial indicators, there are also significant demographic differences among the customer population (male vs. female; education level; and marital status). Frequency analysis showed the diversity of different segments of the customer base. Another finding from the correlation analysis shows that bill amounts across different months are strongly correlated, indicating that customers tend to spend consistently over time. Additionally, credit limits and bill amounts are positively correlated, indicating that customers with a higher credit limit tend to have larger expenditures than those with lower credit limits. Visual representations of some of the financial variables (for example, credit limits and bill amounts) indicate that there are outlier values associated with some financial variables. Outliers may represent customers with unusually high spending patterns. Finally, the analysis shows that customers who defaulted have different credit and repayment histories than customers who did not default, suggesting that

financial behaviors and repayment history are significant variables to consider when developing default risk prediction models. The findings from this exploratory analysis support the conclusion that there is valuable information about customers available through the existing features, which can be leveraged for the purposes of customer segmentation and predictions of future behavior.

6. Conclusions

The analysis completed during EDA gave an overall view into the Data Set for Default of Credit Card Clients. The Data Quality Assessment showed minimal pre-processing was necessary. The Use of Descriptive Statistics and Data Visualization techniques was utilized in the analysis to determine some of the major trends in the customer's demographics, credit limits, spending habits, repayment history, and whether they had defaulted on their credit cards. Correlations between various financial variables were studied as well and trends and clusters of possible outliers were examined through the use of visualizations. The results of this analysis show how EDA can change raw data into actionable information. This analysis will provide a solid base for future applications of Machine Learning and Predictive Analytics regarding assessing credit risk and predicting default of a customer.

AI Usage Disclosure

Generative AI tools were used for coding assistance, conceptual clarification report clarifications and language refinement during the preparation of this assignment. All data analysis, validation of results, interpretation of findings, and conclusions remain the responsibility of the author.